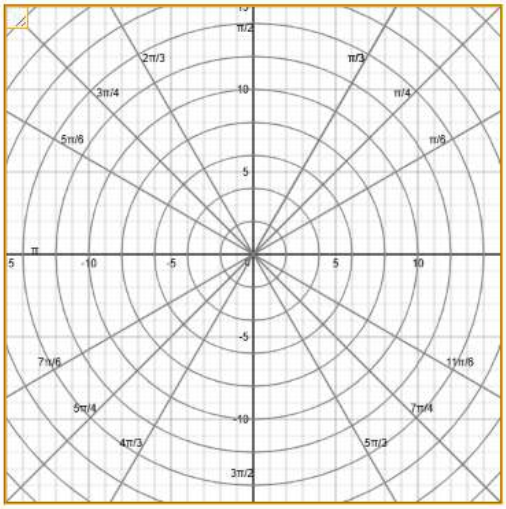
Lesson 34: Conics in Polar Coordinates

1) $r = \dfrac{4}{1 - \cos\theta}$



a) Graph $r = \dfrac{3}{1-\cos\theta}$.

b) eccentricity = __?__

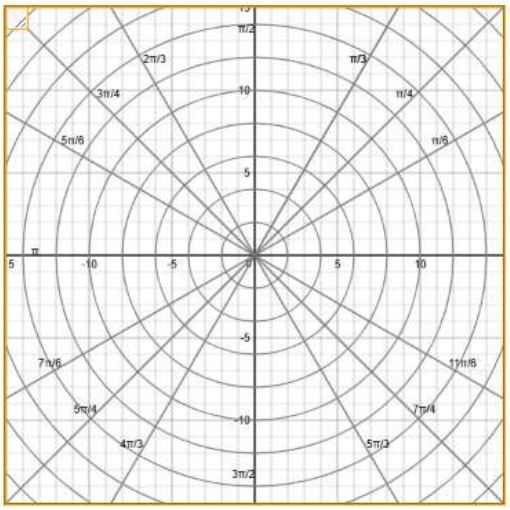
c) type of conic section = __?__ (parabola, ellipse, or hyperbola)

d) Write the corresponding rectangular equation for $r = \dfrac{3}{1-\cos\theta}$.

Hint: Multiply both sides by $1-\cos\theta$.

$y^2 =$ _____?_____

2) Polar equation $r = \dfrac{4}{5-2\cos\theta}$.



a) Graph $r = \dfrac{4}{5-2\cos\theta}$.

b) Multiplying by $\dfrac{1}{5}$: $r = \dfrac{(1/5)\cdot 4}{(1/5)\cdot 5 - (1/5)\cdot 2\cos\theta} =$ __?__

c) eccentricity = __?__

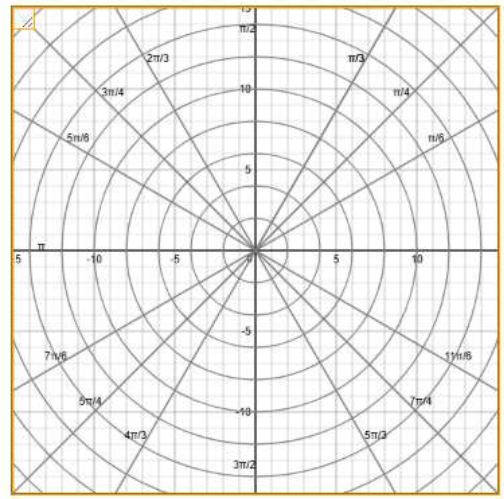
d) Write the corresponding rectangular equation.

Hint: Multiply both sides of $r = \dfrac{4}{5-2\cos\theta}$ by $5-2\cos\theta$.

Use $r = \sqrt{x^2+y^2}$; $x = r\cos\theta$; $y = r\sin\theta$

$25x^2 + 25y^2 =$ _____?_____

3) Polar equation $r = \dfrac{5}{-1 + 4\cos\theta}$.



a) Graph $r = \dfrac{5}{-1 + 4\cos\theta}$.

b) Multiplying by -1: $r = \dfrac{(-1)\cdot 5}{(-1)\cdot(-1) + (-1)\cdot\ 4\cos\theta} =$ ___?___

c) eccentricity = ___?___

d) Write the corresponding rectangular equation.

Hint: Multiply both sides of $r = \dfrac{5}{-1 + 4\cos\theta}$ by $-1 + 4\cos\theta$.

Use $r = \sqrt{x^2 + y^2}$; $x = r\cos\theta$; $y = r\sin\theta$

$x^2 + y^2 =$ _____?_____

4) Find a polar equation for the parabola with its focus at the pole.

Parabola has: $e =$ eccentricity $= 1$; Directrix: $x = -2$

a) focus = _____

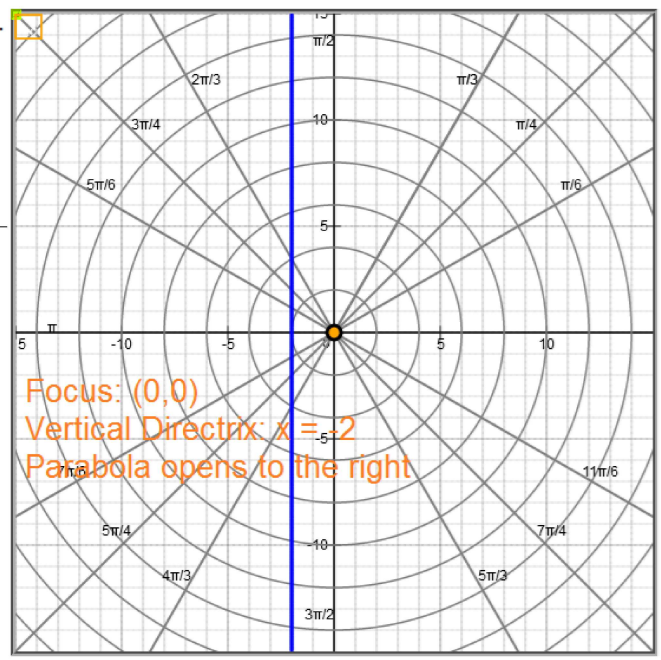
b) vertex = _____

c) Find distance from focus at the pole to directrix $= |d| =$ ___?___

d) Polar equation for parabola: ___?___ (multiple choice)

A) $r = \dfrac{2}{1+\cos\theta}$ B) $r = \dfrac{2}{1-\cos\theta}$

C) $r = \dfrac{-2}{1+\cos\theta}$ D) $r = \dfrac{-2}{1-\cos\theta}$



5) Find a polar equation for the parabola with its focus at the pole. Parabola has vertex at $(1,\ \pi/2)$.

a) focus = _____ b) Equation of directrix: _____

c) Find distance from focus at the pole to directrix $=|d|=$ ___?___

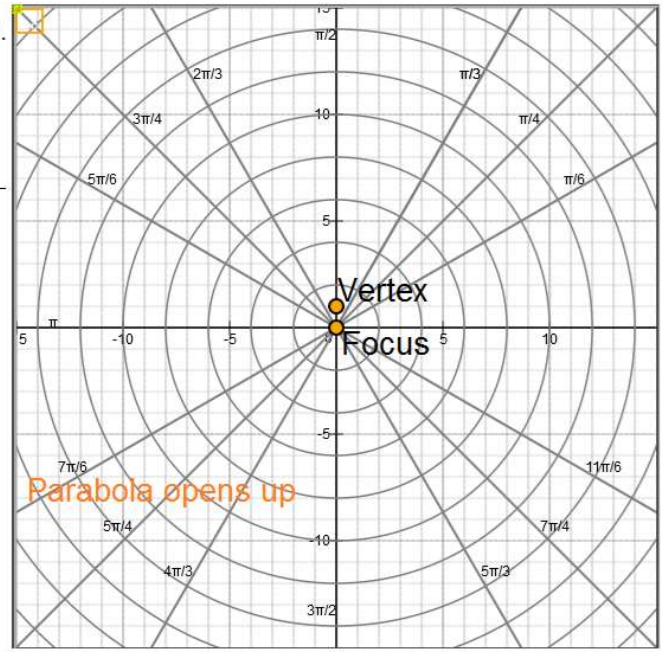
d) e = eccentricity = _____

Hint: To find e, use the equation $r=\dfrac{ed}{1\pm e\sin\theta}$ and the point $(r=1,\ \theta=\pi/2)$.

e) Polar equation for parabola: ___?___ (multiple choice)

A) $r=\dfrac{1}{1+\sin\theta}$ B) $r=\dfrac{2}{1+\sin\theta}$

C) $r=\dfrac{3}{1-\sin\theta}$ D) $r=\dfrac{4}{1-\sin\theta}$



6) Find a polar equation for the ellipse with its focus at the pole. Ellipse has: $e=$ eccentricity $=1/3$; Directrix: $y=3$

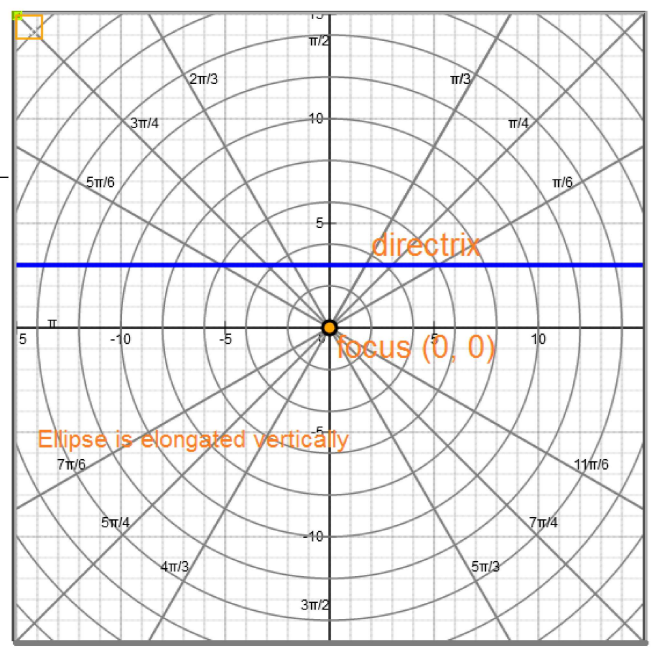
a) focus =_____

b) Find distance from focus at the pole to directrix $=|d|=$ ___?___

c) Polar equation for ellipse: ___?___

Equation of ellipse will be: $r=\dfrac{ed}{1+e\sin\theta}$ or $r=\dfrac{ed}{1-e\sin\theta}$.

Note: One of the vertices of the ellipse is closer to the pole. Directrix should be near the vertex closest to the pole.



7) Find a polar equation for the ellipse with its focus at the pole.
Ellipse has vertices at (2, 0) and (6, π).

a) focus = ___?___

b) center of ellipse = ___?___

c) $a =$ distance from center to vertex = ___?___

d) $c =$ distance from center to focus = ___?___

e) $e =$ eccentricity $= c/a =$ ___?___

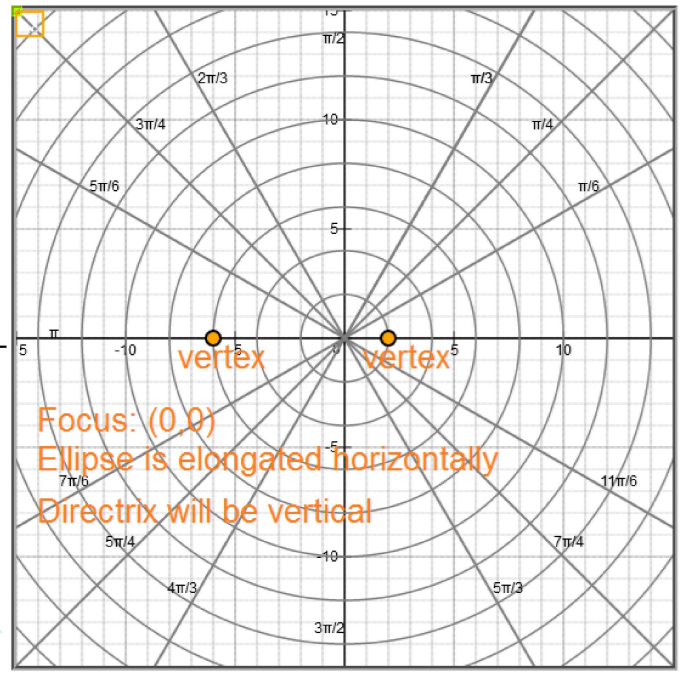
f) Find distance from focus at the pole to directrix $= |d| =$ ___?___

Hint: Equation of ellipse will have the form: $r = \dfrac{ed}{1 \pm e\cos\theta}$

Use the point ($r = 2$, $\theta = 0$) to find d.

g) Polar equation for ellipse: ___?___

Note: One of the vertices of the ellipse is closer to the pole.
Directrix should be near the vertex closest to the pole.



8) Find a polar equation for the hyperbola with its focus at the pole.

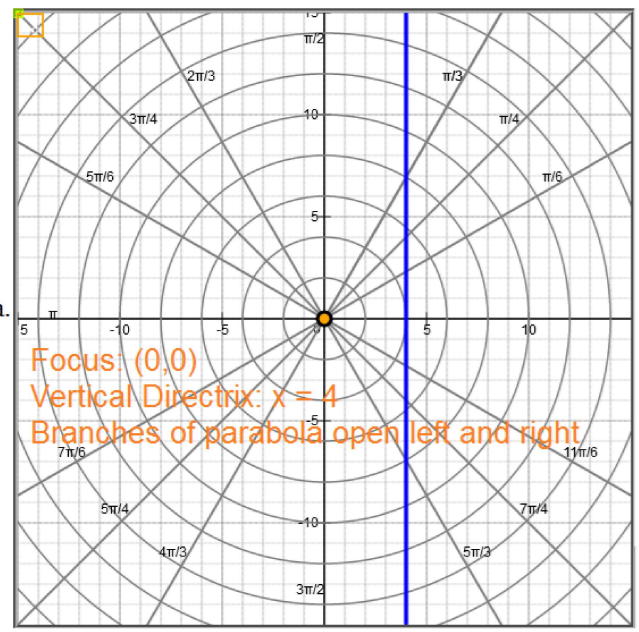
Hyperbola has: $e =$ eccentricity $= 4$; Directrix: $x = 4$

a) Find distance from focus at the pole to directrix $= d =$ ___?___

b) Polar equation for hyperbola: ___?___

Equation of hyperbola will be: $r = \dfrac{ed}{1 + e\cos\theta}$ or $r = \dfrac{ed}{1 - e\cos\theta}$.

Note: Directrix should be between the two branches of the hyperbola.



9) Find a polar equation for the hyperbola with its focus at the pole.

Hyperbola has vertices at $(1,\ 3\pi/2)$ and $(7,\ 3\pi/2)$.

a) focus = ___?___

b) center of hyperbola = ___?___

c) $a =$ distance from center to vertex = ___?___

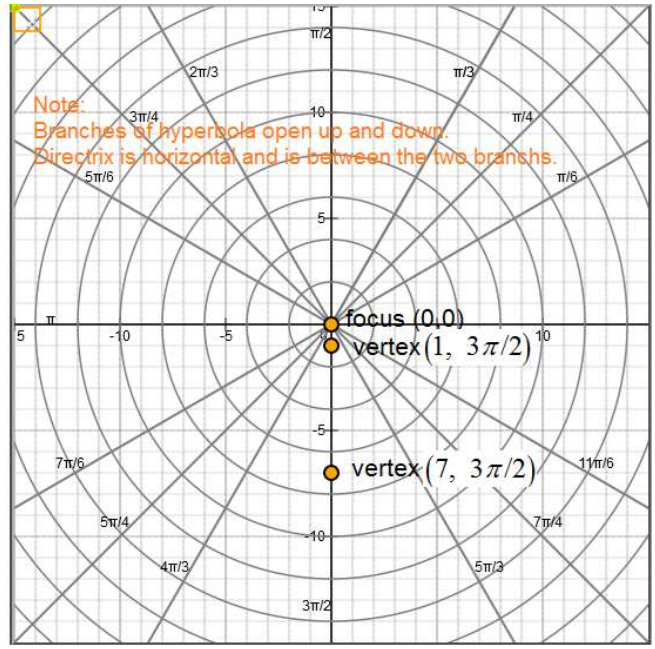
d) $c =$ distance from center to focus = ___?___

e) $e =$ eccentricity $= c/a =$ ___?___

f) Find distance from focus at the pole to directrix $= |d| =$ ___?___

Hint: Equation of hyperbola will have the form: $r = \dfrac{ed}{1 \pm e\sin\theta}$.

Use the point $(r = 1,\ \theta = 3\pi/2)$ to find d.

g) Polar equation for hyperbola: ___?___

Note: Directrix should be between the two branches of the hyperbola.